

Spatial Economics - TA Session 4

Caliendo, Dvorkin, and Parro (2019)

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I thank Daniil Iurchenko as these slides build on his original slides. All errors are mine.

February 20, 2026

Plan for today

- Goal: cover details of CDP (2019) and discuss its numerical implementation
- Plan for today:
 - 1 Recap CDP model
 - 2 Discuss the procedure for counterfactuals and underlying assumptions
 - 3 Discuss the numerical implementation

Outline of the model

- N locations and J sectors
- Costs of trading goods: iceberg trade costs $\kappa_t^{nj,ij}$
- **Costs of moving people**: additive labor reallocation costs $\tau^{nj,ik}$
- Preferences, production, and market structure:
 - Similar to Caliendo, Parro, Rossi-Hansberg, and Sarte (2018)
- **Household's problem**: choose a history of location-sector pairs that maximizes expected PDV of utility given (a) history of iid location-sector idiosyncratic shocks, (b) labor reallocation costs, and (c) path of the economy (i.e., perfect foresight)

Fundamentals and Parameters

- Time-constant fundamentals: $\overline{\Theta} = (Y, H, b)$
 - Y : a matrix of labor reallocation costs
 - H : a vector of location-specific structures
 - b : a vector of location-specific home production technologies
- Time-varying fundamentals: $\Theta_t = (A_t, \kappa_t)$
 - A_t : a matrix of location-sector-specific productivities
 - κ_t : a matrix of iceberg trade costs
- Parameters:
 - Preferences: expenditure shares α^j , discount factor β , and migration elasticity ν
 - Production: value-added shares γ^{nj} , labor shares $1 - \zeta^n$ in value added, input-output coefficients $\gamma^{nj,ik}$, and trade elasticity θ

Temporary Equilibrium

TEMPORARY EQUILIBRIUM. Given $(L_t, \Theta_t, \bar{\Theta})$, a temporary equilibrium is a vector of wages $w(L_t, \Theta_t, \bar{\Theta})$ that satisfies the equilibrium conditions of the static subproblem.

Remarks.

- No labor mobility in the static subproblem
- No need to show equations since they are similar to Caliendo et al (2018)
- Can solve for the equilibrium wages and rents using the algorithm from Caliendo and Parro (2015) or Caliendo et al (2018)
- We do not need rents for the sequential equilibrium

Sequential Competitive Equilibrium

SEQUENTIAL COMPETITIVE EQUILIBRIUM. Given $(L_t, \Theta_t, \bar{\Theta})$, a sequential competitive equilibrium is a sequence of $\{L_t, \mu_t, V_t, w(L_t, \Theta_t, \bar{\Theta})\}$ that solves the system

$$V_t^{nj} = U(C_t^{nj}) + \nu \log \left(\sum_{i=1}^N \sum_{k=0}^J \exp(\beta V_{t+1}^{ik} - \tau^{nj,ik})^{1/\nu} \right) \quad (1)$$

$$\mu_t^{nj,ik} = \frac{\exp(\beta V_{t+1}^{ik} - \tau^{nj,ik})^{1/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \exp(\beta V_{t+1}^{mh} - \tau^{nj,mh})^{1/\nu}} \quad (2)$$

$$L_{t+1}^{nj} = \sum_{i=1}^N \sum_{k=0}^J \mu_t^{ik,nj} L_t^{ik} \quad (3)$$

as well as temporary equilibrium at each period t .

Data

- Data: $(\pi_t, w_t L_t + r_t H_t, L_t, \mu_t)$ for 2000–2007
 - Initial period $t = 0$: year 2000
 - Initial conditions for the model: $(L_0, \pi_0, X_0, \mu_{-1})$
 - X_0 constructed from π_0 and value added
 - Value added $(w_t L_t + r_t H_t)_{t=0}$ used to calibrate parameters: $\gamma^{nj}, \gamma^{nj,nk}, \zeta^n, \alpha^j, \iota^n$
- Locations: 50 US states + 37 countries (incl. China) + rest of world
- Sectors: 22 sectors from NAICS (similar to CP 2015)
- To get trade shares between US states and countries, assume that the share of each state in total US trade with any country is determined by regional share of employment in an industry
- See appendix of the paper for a detailed description of data sources

Calibration

- Calibrate α^j , γ^{nj} , ξ^n , $\gamma^{nj,ik}$, and ι^n using data from BEA and WIOD
- Set $\beta = 0.99$
- Take trade elasticities θ^j from Caliendo and Parro (2015)
- To estimate migration elasticity ν , use the model-implied equation

$$\log \left(\mu_t^{nj,nk} / \mu_t^{nj,nj} \right) = \tilde{C} + \frac{\beta}{\nu} \log \left(w_{t+1}^{nk} / w_{t+1}^{nj} \right) + \beta \log \left(\mu_{t+1}^{nj,nk} / \mu_{t+1}^{nj,nj} \right) + \omega_{t+1}$$

- Use log utility and equilibrium equations for V_t^{nj} and $\mu_t^{nj,nk}$
- $\tilde{C}$ is a constant, ω_{t+1} is a random term that is correlated with the regressors
- Use lagged migration flows and wages as instruments

Baseline Economy

BASILINE ECONOMY. Baseline economy is the allocation $\{L_t, \mu_{t-1}, \pi_t, X_t\}_{t=0}^{\infty}$ corresponding to the sequence of fundamentals $\{\Theta_t\}_{t=0}^{\infty}$ and $\bar{\Theta}$.

Remarks.

- Agents have perfect foresight about the sequence of fundamentals $\{\Theta_t\}_{t=0}^{\infty}$ and know how the economy would evolve
- Before we study counterfactuals, we need to solve for the baseline economy
→ First application of dynamic hat algebra

Solving for Baseline Economy: Remarks (1/2)

- Trick similar to static hat algebra – rewrite the equilibrium conditions in changes
 - Rewriting the static equilibrium conditions in changes is standard hat algebra
 - We need to rewrite the forward-looking equations in changes
- Why does it work?
 - Model is log-linear
 - Enough to recover Θ_0 and $\bar{\Theta}$ since the system is forward-looking
 - Given (L_0, μ_{-1}) , data on π_0 and X_0 provides enough information about fundamentals (productivity, etc.) to be able to compute changes in static variables
 - Data on (L_0, μ_{-1}) provides enough information about migration costs to be able to compute changes in dynamic variables

Solving for Baseline Economy: Remarks (2/2)

- Note that to compute the baseline equilibrium, we **have to know** the whole path of changes in fundamentals
- Dynamic exact hat algebra provides information on fundamentals $(\Theta_0, \bar{\Theta})$ **conditional** on our guess of $\{\dot{\Theta}_t\}_{t=1}^{\infty}$ being correct
 - Responses of agents that are described by $(L_0, \mu_{-1}, \pi_0, X_0)$ include the anticipation effects
- Note that the sequence of changes $\{\dot{\Theta}_t\}_{t=1}^{\infty}$ have to be **convergent**
 - In theory, we can consider any sequence, but we would not be able to numerically solve for the equilibrium

Algorithm

Initialize algorithm with:

- 1 Initial allocation: $L_0, \pi_0, X_0, \mu_{-1}$
- 2 A convergent path of changes in fundamentals $\{\dot{\Theta}_t\}_{t=1}^T$ with $\dot{\Theta}_T \approx 1$
- 3 A convergent path for value functions $\{\dot{u}_{t+1}^{(0)}\}_{t=0}^T, \dot{u}_T^{(0)} \approx 1$

SHOOTING ALGORITHM

- 1 Simulate forward using $\{\dot{u}_{t+1}^{(i)}\}_{t=0}^T$ to compute $\{\mu_{t+1}\}_{t=0}^T$

$$\mu_{t+1}^{nj,ik} = \frac{\mu_t^{nj,ik} (\dot{u}_{t+2}^{ik})^{\beta/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \mu_t^{nj,mh} (\dot{u}_{t+2}^{mh})^{\beta/\nu}}$$

2. Simulate forward using $\{\mu_{t+1}\}_{t=0}^T$ to compute $\{L_{t+1}\}_{t=0}^T$

$$L_{t+1}^{nj} = \sum_{i=1}^N \sum_{k=0}^J \mu_t^{ik,nj} L_t^{ik}$$

3. Solve the temporary equilibrium for each t to get $\{\dot{w}_{t+1}, \dot{P}_{t+1}\}_{t=0}^T$

4. Compute $\{\dot{u}_{t+1}^{(i+1)}\}_{t=0}^T$ by solving the system backwards using updated $\dot{w}_{t+1}$, $\dot{P}_{t+1}$, and μ_t

$$\dot{u}_{t+1}^{nj} = \frac{\dot{w}_{t+1}^{nj}}{\dot{P}_{t+1}^n} \left(\sum_{i=1}^N \sum_{k=0}^J \mu_t^{nj,ik} \left(\dot{u}_{t+2}^{ik} \right)^{\beta/\nu} \right)^\nu$$

5. Check whether $\sum_{t=0}^T \|\dot{u}_{t+1}^{(i+1)} - \dot{u}_{t+1}^{(i)}\| < \epsilon$. If not, take $\{\dot{u}_{t+1}^{(i+1)}\}_{t=0}^T$ as the new guess and go to step $i + 1$.

Algorithm: Remarks

- Equations for the temporary equilibrium
 - Similar to Caliendo and Parro (2015) and Caliendo et al (2018)
 - But instead of exogenous $\dot{L}_t$, we use the equilibrium at $t - 1$ to compute $\dot{L}_t$
- How to pick an initial guess for $\dot{u}_{t+1}^{(0)}$?
 - Setting $\dot{u}_{t+1}^{(0)} = 1$ should work fine
- Some other ideas to try
 - Make $\dot{u}_{t+1}^{(0)}$ proportional to $\dot{A}_t$
 - Use the exact hat algebra to solve a sequence of static equilibria

Solving for Counterfactual Equilibrium: Remarks

- Counterfactuals: deviations from the baseline allocation
 - Intuition is similar to difference in differences
 - We compute counterfactuals **conditional** on our guess of $\{\dot{\Theta}_t\}_{t=1}^{\infty}$ being correct
- Assumption about households' **expectations**
 - At period $t = 0$, agents anticipate $\{\Theta_t\}_{t=1}^{\infty}$
 - There is no anticipation of changes in the path of fundamentals
 - At period $t = 1$, agents learn about the entire counterfactual sequence $\{\Theta'_t\}_{t=1}^{\infty}$
- Similar to MIT shock

Dynamic Hat Algebra: Outline

- Notation:
 - Time changes for the baseline economy: $\dot{y}_{t+1} = y_{t+1}/y_t$
 - Time changes for the ctf equilibrium: $\dot{y}'_{t+1} = y'_{t+1}/y'_t$
 - Deviation of ctf equilibrium from the baseline: $\hat{y}_{t+1} = \dot{y}'_{t+1}/\dot{y}_{t+1}$
- Fundamentals $\overline{\Theta}$ stay the same and are “calibrated” by the initial condition
 - Rewrite the system in deviations from the baseline equilibrium
 - The system is similar, except two new objects: $\pi_t'^{nj,ij} \dot{\pi}_{t+1}^{nj,ij}$ and $\kappa_t'^{nj,ij} \dot{\kappa}_{t+1}^{nj,ij}$
 - These combine the counterfactual level at t with the baseline change at $t + 1$
- Intuition for $\pi_t'^{nj,ij} \dot{\pi}_{t+1}^{nj,ij}$ is similar to DiD
 - $\pi_t'^{nj,ij}$ gives information about differences in levels
 - $\dot{\pi}_{t+1}^{nj,ij}$ capture the “common trend”

Algorithm: Sequential Equilibrium

Initialize algorithm with:

- 1 Baseline allocation: $\{L_t, \mu_t, \pi_t, X_t\}_{t=0}^T$
- 2 A counterfactual convergent path of changes in fundamentals: $\{\hat{\Theta}_t\}_{t=0}^T$, $\hat{\Theta}_T \approx 1$
- 3 A convergent path for value functions $\{\hat{u}_{t+1}^{(0)}\}$, $\hat{u}_0^{(0)} = 1$, $\hat{u}_T^{(0)} \approx 1$

SHOOTING ALGORITHM

- 1 Simulate forward using $\{\hat{u}_{t+1}^{(i)}\}_{t=0}^T$ to compute $\{\mu'_{t+1}\}$:

$$\mu_t'^{nj,ik} = \begin{cases} \mu_0^{nj,ik} & \text{if } t = 0 \\ \frac{\mu_1^{nj,ik} (\hat{u}_1^{ik})^{\beta/\nu} (\hat{u}_2^{ik})^{\beta/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \mu_1^{nj,mh} (\hat{u}_1^{mh})^{\beta/\nu} (\hat{u}_2^{mh})^{\beta/\nu}} & \text{if } t = 1 \\ \frac{\mu_{t-1}'^{nj,ik} \dot{\mu}_t^{nj,ik} (\hat{u}_{t+1}^{ik})^{\beta/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \mu_{t-1}'^{nj,mh} \dot{\mu}_t^{nj,mh} (\hat{u}_{t+1}^{mh})^{\beta/\nu}} & \text{if } t > 1 \end{cases}$$

2. Simulate forward using $\{\mu'_{t+1}\}$ to compute $\{L'_{t+1}\}$:

$$L'^{nj}_{t+1} = \sum_{i=1}^N \sum_{k=0}^J \mu'^{ik,nj}_t L'^{ik}_t$$

3. Solve the temporary equilibrium for each t to get $\{\hat{w}_{t+1}, \hat{P}_{t+1}\}$
4. Compute $\{\hat{u}^{(i+1)}_{t+1}\}_{t=0}^T$ by solving the system backwards using updated $\hat{w}_{t+1}$, $\hat{P}_{t+1}$, μ'_t , and $\dot{\mu}_t$

$$\hat{u}^{nj}_t = \begin{cases} \left(\frac{\hat{w}^{nj}_t}{\hat{P}^n_t} \right) \left(\sum_{i=1}^N \sum_{k=0}^J \mu'^{nj,ik}_{t-1} \dot{\mu}^{nj,ik}_t \left(\hat{u}^{ik}_{t+1} \right)^{\beta/\nu} \right)^\nu & \text{for } t \geq 2 \\ \left(\frac{\hat{w}^{nj}_1}{\hat{P}^n_1} \right) \left(\sum_{i=1}^N \sum_{k=0}^J \mu_1^{nj,ik} \left(\hat{u}^{ik}_1 \right)^{\beta/\nu} \left(\hat{u}^{ik}_2 \right)^{\beta/\nu} \right)^\nu & \text{for } t = 1 \end{cases}$$

5. Check whether $\sum_{t=0}^T \|\hat{u}^{(i+1)}_{t+1} - \hat{u}^{(i)}_{t+1}\| < \epsilon$. If not, take $\{\hat{u}^{(i+1)}_{t+1}\}_{t=0}^T$ as the new guess and go to step $i + 1$.

Replication Package

- Base_Year/Base_year.m
 - Compute initial equilibrium allocations for year 2000 from raw data (trade flows, IO tables, gross output, value added shares) and model equilibrium conditions
- time_varying_fundamentals/Baseline_economy/Step_1_data.m
 - Construct time series data for 2000-2007: bilateral trade flows, trade shares, expenditures, changes in wages, labor allocations, and evolution of gross flows
- time_varying_fundamentals/Baseline_economy/Step_2_Baseline_00_07.m
 - Solve temporary equilibrium at each t for 2000-2007 using Step 1 data and Base Year allocations
- time_varying_fundamentals/Baseline_economy/Step_3_Baseline_2007_forward.m
 - Solve 2007 forward with constant fundamentals using baseline shooting algorithm
- time_varying_fundamentals/Baseline_economy/Step_4_Baseline.m
 - Merge 2000-2007 actual allocations with 2007+ simulated allocations to create the baseline economy
- time_varying_fundamentals/Counterfactual_economy/Counterfactual_economy.m
 - Solve counterfactual: what would happen absent the China productivity shock (all fundamentals as in baseline except Chinese productivities)